

1) Explain all conditional statements with examples

Conditional statements are used when we want Python to **make a decision** based on a condition.

Python mainly has these conditional statements:

1. if
2. if...else
3. if...elif...else
4. Nested if

1. if Statement

The if statement is used when we want to execute some code **only when a condition is true**.

Syntax

```
if condition:  
    statement
```

Example

```
age = 20  
if age >= 18:  
    print("You are eligible to vote")
```

Output

You are eligible to vote

2. if...else Statement

The if...else statement is used when there are **two possible choices**.

- If the condition is true - execute if
- If the condition is false - execute else

Syntax

```
if condition:
    statement1
else:
    statement2
```

Example

```
age = 15
if age >= 18:
    print("You can vote")
else:
    print("You cannot vote")
```

Output

You cannot vote

3. if...elif...else Statement

This is used when we have **multiple conditions**.

- if - checks the first condition
- elif - checks another condition
- else - runs when none of the conditions are true

Syntax

```
if condition1:
    statement1
elif condition2:
    statement2
elif condition3:
    statement3
else:
```

statement4

Example

```
marks = 75
```

```
if marks >= 90:
```

```
    print("Grade A+")
```

```
elif marks >= 75:
```

```
    print("Grade A")
```

```
elif marks >= 60:
```

```
    print("Grade B")
```

```
else:
```

```
    print("Grade C")
```

Output

Grade A

4. Nested if Statement

A **nested if** means having one if statement **inside another if statement**.

Example

```
age = 20
```

```
citizen = True
```

```
if age >= 18:
```

```
    if citizen:
```

```
        print("You can vote")
```

```
    else:
```

```
        print("You must be a citizen")
```

```
else:
```

```
    print("You are under 18")
```

Output

You can vote